



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 01



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Recap of Previous Lecture



Topic

Basic Formulas



Topics to be Covered



Topic

Basic of Graph





Topic : Basic of Graph

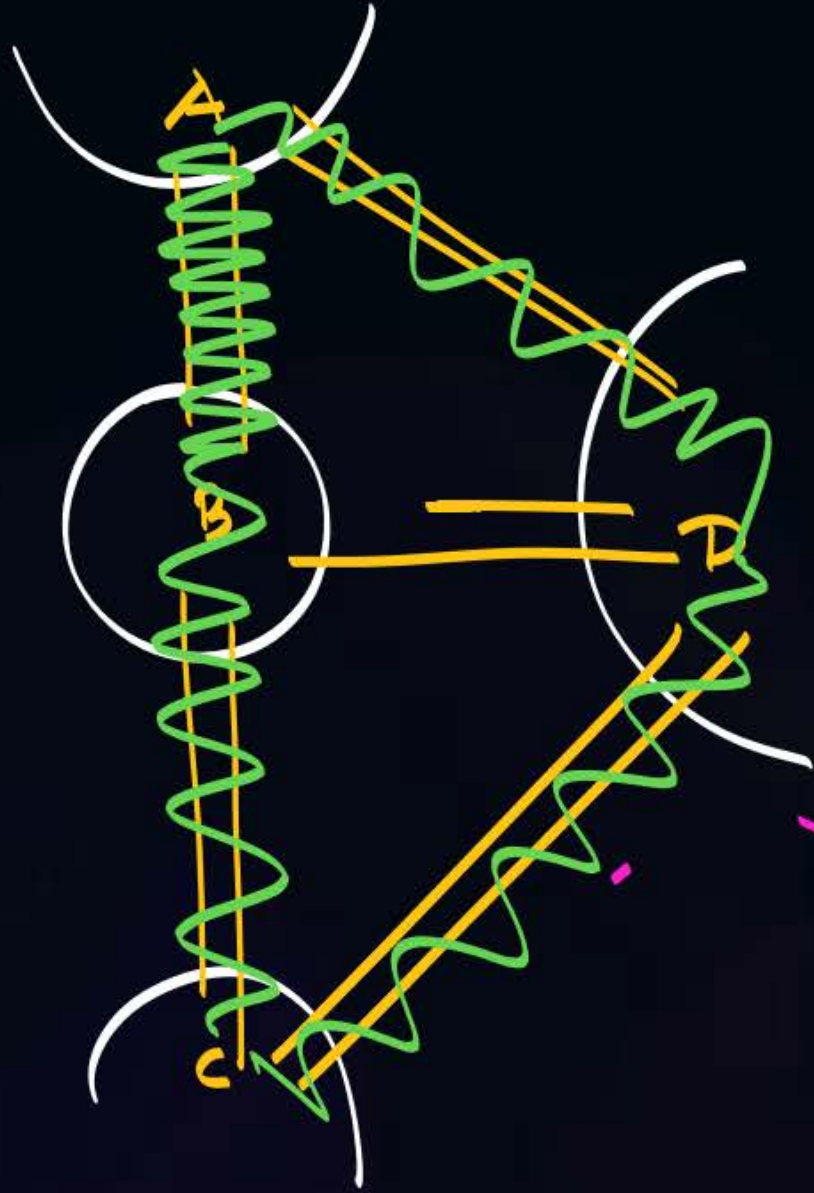


Dm : (10+

- 1) Graph (4 - 6)
- 2) logic (2 - 4)
- 3) Set theory (2 - 4)
- 4) Combinatorics (2 - 4)



Topic : Basic of Graph



- 1) A 
- 2) all bridges exactly ones.

✓	✗
✗	✓

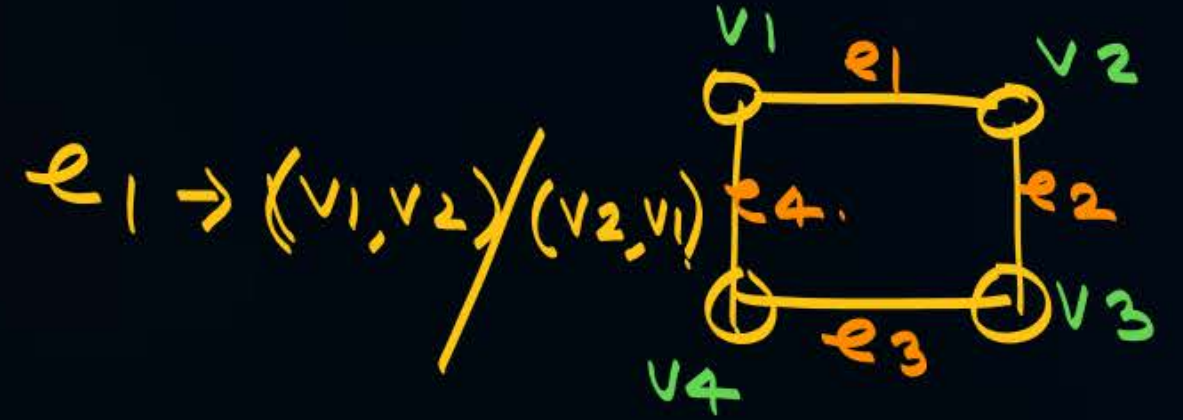




Topic : Basic of Graph

point/joint $\rightarrow$ vertex/vertices.

line/branch $\rightarrow$ edge/edges.



Graph $G = (V, E)$

set of
vertices.

set of
edges.

$G = (V, E)$

$V = \{v_1, v_2, v_3, v_4\}$

$E = \{e_1, e_2, e_3, e_4\}$

$V =$ set of vertices $= \{v_1, v_2, \dots, v_n\}$

$E =$ set of edges $= E = \{e_1, e_2, e_3, \dots, e_n\}$

each edge must be associated with unordered pair of vertices.



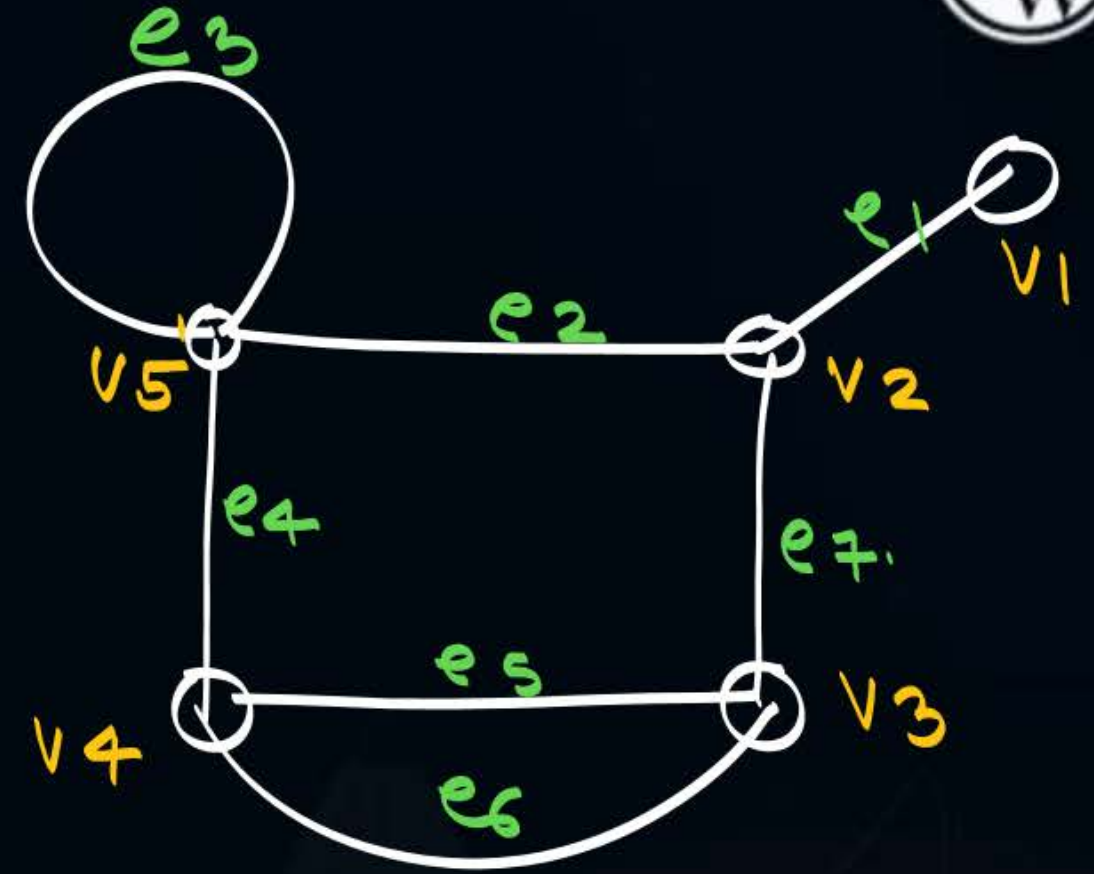
Topic : Basic of Graph

end vertices :

edge $\rightarrow$ unordered pair of vertices $\rightarrow$ end vertices. $e_1 \rightarrow (v_1, v_2)$

loop / self-loop :

end vertices are same (edge $\rightarrow$ loop.)



$e_3 \rightarrow (v_5, v_5)$

|| edges :

$e_5 \rightarrow (v_4, v_3)$

$e_6 \rightarrow (v_4, v_3)$

2 or more edges having same end vertices then they are || edges



Topic : Basic of Graph



incident point :

meeting point of vertex & edge.

Degree/valency :

no. of incident point.

pendant vertex.

Degree 1 vertex.
is pendant vertex.

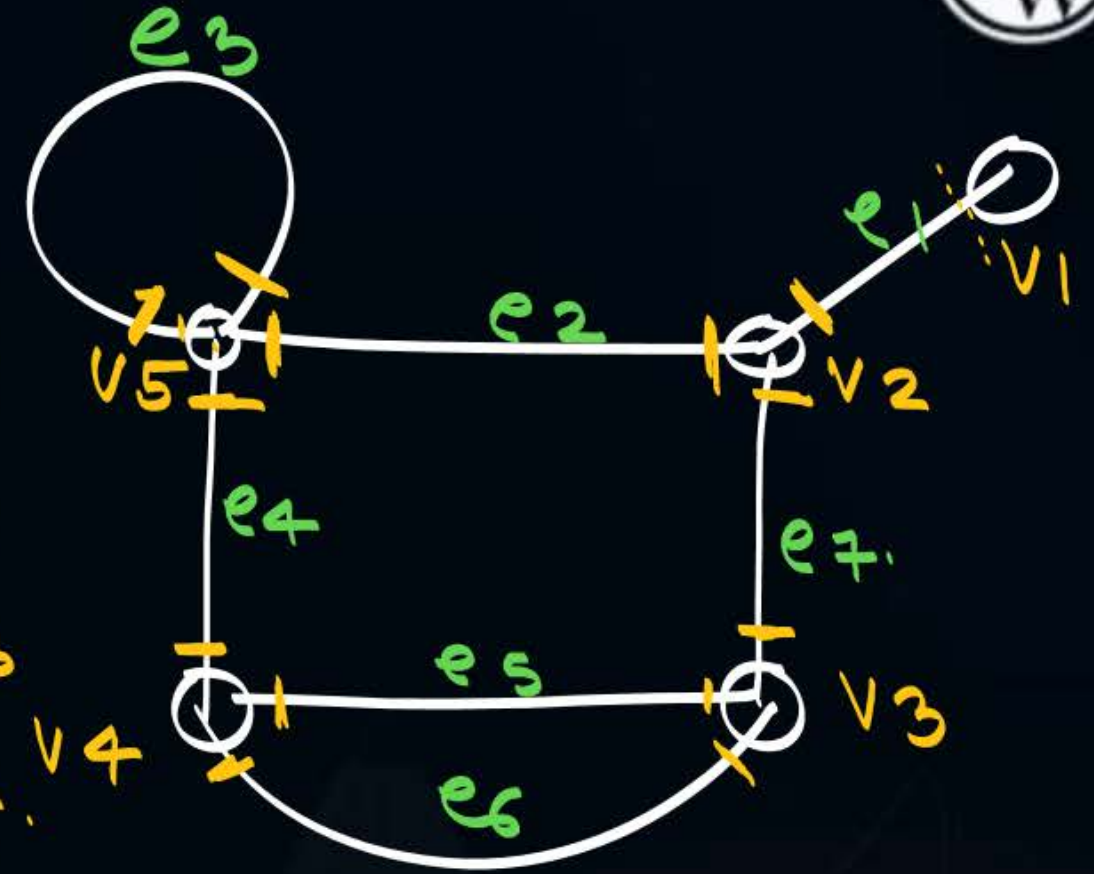
$$d(v_1) = 1$$

$$d(v_2) = 3$$

$$d(v_3) = 3$$

$$d(v_4) = 3$$

$$d(v_5) = 4$$



isolated vertex :

Degree 0 vertex is isolated vertex.



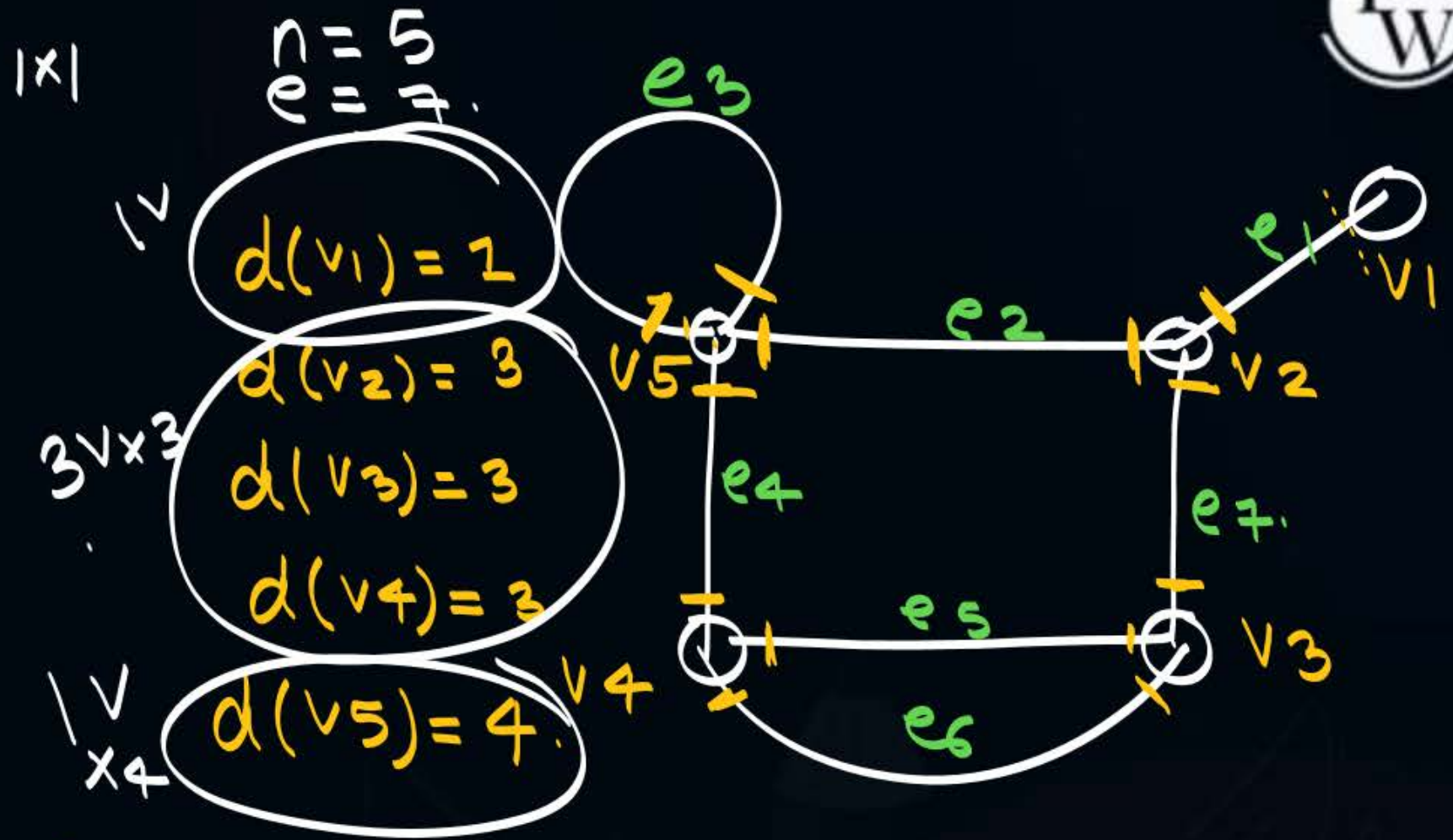
Topic : Basic of Graph



$$d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5)$$

$$= 1 + 3 + 3 + 3 + 4$$

$$= 14 = 2 \times 7$$



Thm 1:

Sum of degrees of all vertices is equals to twice the no. of edges.

$$\sum d(v_i) = 2e$$

L.H.S R.H.S.
 Degrees edges.

$$2 = 2 \times 1$$

$$2 + 2 = 2(1 + 1)$$

$$2 + 2 + 2 = 2(1 + 1 + 1)$$





Trivial Graph.

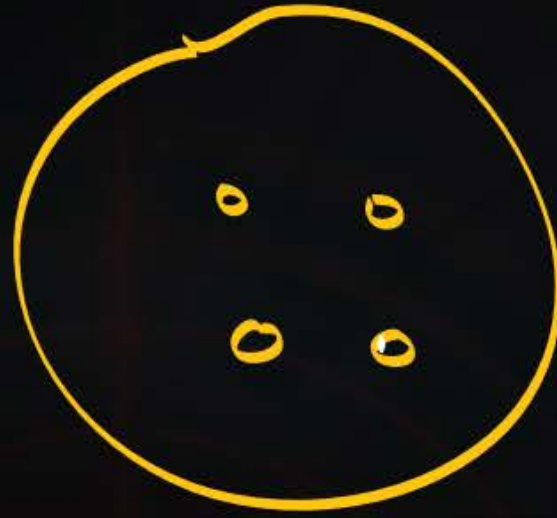
$$G = (V, E)$$

↓
set

↓
set

$$V \neq \emptyset$$

$$E = \emptyset$$



Null Graph

Set of isolated vertices.



Topic : Basic of Graph

Graph is having

$$n = 14 \quad e = 27$$

Degree of each vertex is

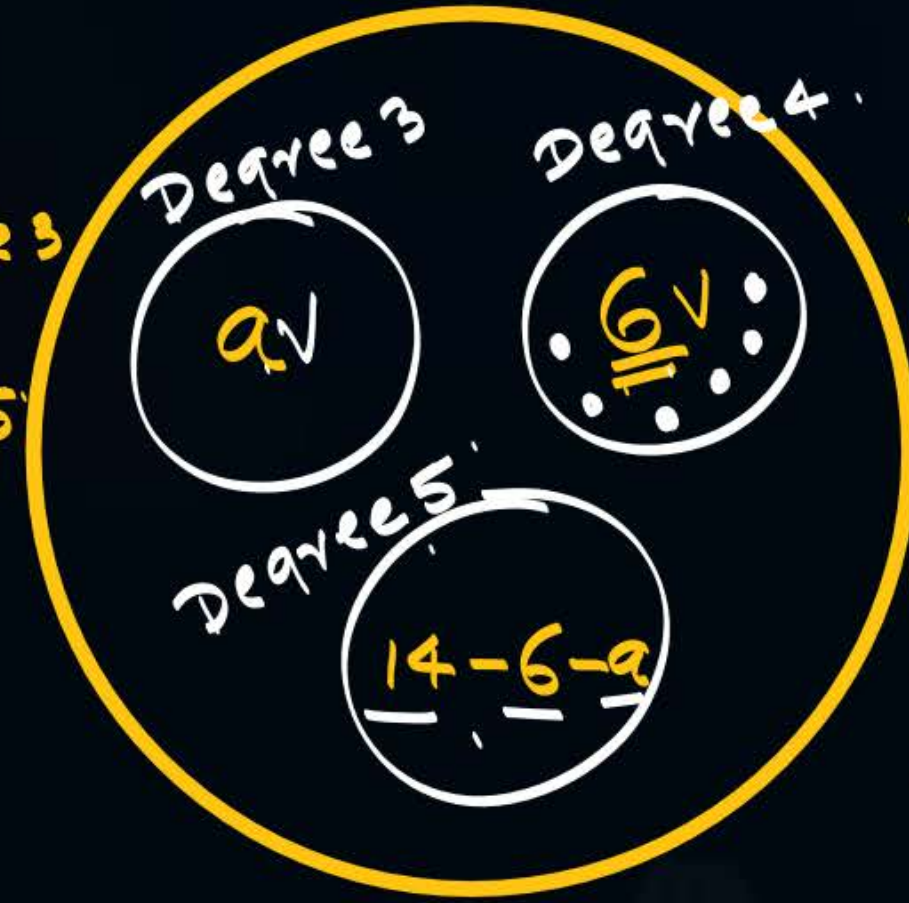
3, 4 or 5

There 6 vertices of Degree 4.

how many vertices G have Degree 3

& 5?

$\left\{ \begin{array}{l} 5v \text{ of Degree 3} \\ 3v \text{ of Degree 5} \end{array} \right.$



$$\frac{n = 14}{e = 27}$$

Assume
a vertices
of Degree 3.

$$\sum d(v_i) = 2e$$

$$\underline{6 \times 4} + \underline{3a} + 5(14 - 6 - a) = 2 \cdot 27$$

$$24 + 3a + 5(8 - a) = 54$$

$$24 + 3a + 40 - 5a = 54$$

$$a = 5$$

$$n = 25 \quad e = 62.$$

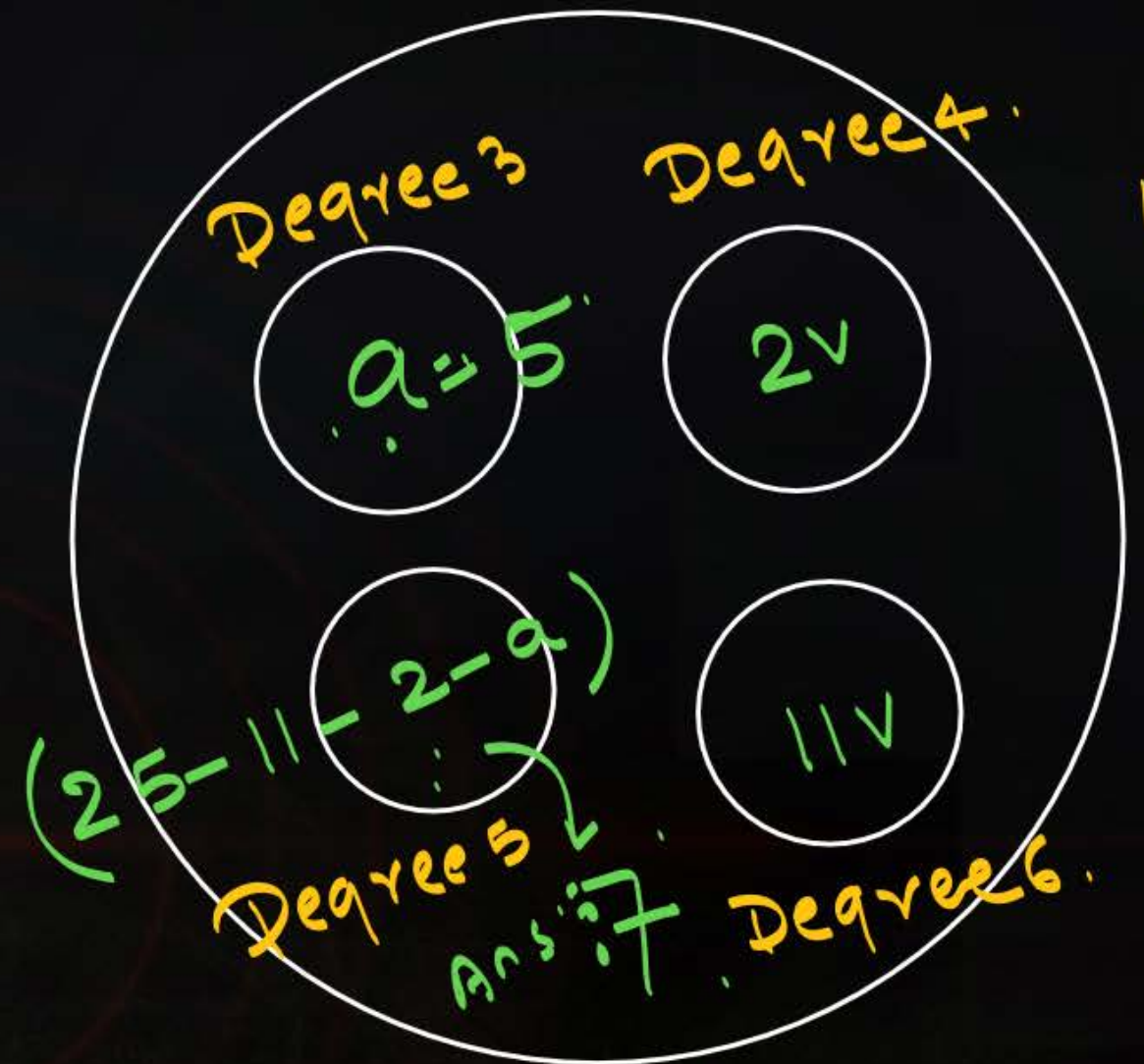
Degree of each vertex is
3, 4, 5 or 6.

2 vertices of Degree 4.

11 vertices of Degree 6.

how many vertices of Degree 5?

$$\begin{aligned} \text{Ans: } (25 - 11 - 2 - a) \\ = (12 - a) \\ = 12 - 5 = 7 \end{aligned}$$



$$\begin{aligned} n &= 25 \\ e &= 62. \end{aligned}$$

$$\begin{aligned} 3 \cdot a + 4 \cdot 2 + (25 - 11 - 2 - a) \times 5 + 11 \times 6 \\ = 262. \end{aligned}$$

$$a = 5$$



2 mins Summary

Topic

One

Basic of Graph

Topic

Two

Topic

Three

Topic

Four

Topic

Five

THANK - YOU